

5.8 Exercises

Exercise 5.1: Hausdorff distance between a set and a subset

Show that $d_H(\mathbb{A}, \mathbb{B}) = \max_{a \in \mathbb{A}} d(a, \mathbb{B})$ if $\mathbb{A}$ and $\mathbb{B}$ are two compact subsets of $\mathbb{R}^n$ satisfying $\mathbb{B} \subseteq \mathbb{A}$.

Exercise 5.2: Hausdorff distance between sets $\mathbb{A} \oplus \mathbb{B}$ and $\mathbb{B}$

Show that $d_H(\mathbb{A} \oplus \mathbb{B}, \mathbb{A}) \leq |\mathbb{B}|$ if $\mathbb{A}$ and $\mathbb{B}$ are two compact subsets of $\mathbb{R}^n$ satisfying $0 \in \mathbb{B}$ in which $|\mathbb{B}| := \max_b \{|b| \mid b \in \mathbb{B}\}$.

Exercise 5.3: Hausdorff distance between sets $\{z\} \oplus \mathbb{B}$ and $\mathbb{A}$

Show that $d_H(\{z\} \oplus \mathbb{B}, \mathbb{A}) \leq |z| + d_H(\mathbb{A}, \mathbb{B})$ if $\mathbb{A}$ and $\mathbb{B}$ are two compact sets in $\mathbb{R}^n$.

Exercise 5.4: Hausdorff distance between sets $\{z\} \oplus \mathbb{A}$ and $\mathbb{A}$

Show that $d_H(\{z\} \oplus \mathbb{A}, \mathbb{A}) = |z|$ if z is a point and $\mathbb{A}$ is a compact set in $\mathbb{R}^n$.

Exercise 5.5: Hausdorff distance between sets $\mathbb{A} \oplus \mathbb{C}$ and $\mathbb{B} \oplus \mathbb{C}$

Show that $d_H(\mathbb{A} \oplus \mathbb{C}, \mathbb{B} \oplus \mathbb{C}) = d_H(\mathbb{A}, \mathbb{B})$ if $\mathbb{A}$, $\mathbb{B}$, and $\mathbb{C}$ are compact subsets of $\mathbb{R}^n$ satisfying $\mathbb{B} \subseteq \mathbb{A}$.

Exercise 5.6: Hausdorff distance between sets $F\mathbb{A}$ and $F\mathbb{B}$

Let $\mathbb{A}$ and $\mathbb{B}$ be two compact sets in $\mathbb{R}^n$ satisfying $\mathbb{A} \subseteq \mathbb{B}$, and let $F \in \mathbb{R}^{n \times n}$. Show that $d_H(F\mathbb{A}, F\mathbb{B}) \leq |F| d_H(\mathbb{A}, \mathbb{B})$ in which $|F|$ is the induced norm of F satisfying $|Fx| \leq |F| |x|$ and $|x| := d(x, 0)$.

Exercise 5.7: Linear combination of sets; $\lambda_1 \mathbb{W} \oplus \lambda_2 \mathbb{W} = (\lambda_1 + \lambda_2) \mathbb{W}$

If $\mathbb{W}$ is a convex set, show that $\lambda_1 \mathbb{W} \oplus \lambda_2 \mathbb{W} = (\lambda_1 + \lambda_2) \mathbb{W}$ for any $\lambda_1, \lambda_2 \in \mathbb{R}_{\geq 0}$. Hence show $\mathbb{W} \oplus \lambda \mathbb{W} \oplus \lambda^2 \mathbb{W} \oplus \dots = (1 - \lambda)^{-1} \mathbb{W}$ if $\lambda \in [0, 1)$.

Exercise 5.8: Hausdorff distance between the sets $\Phi(i)$ and Φ

Show that there exist $c > 0$ and $\gamma \in (0, 1)$ such that

$$d_H(\Phi(i), \Phi) \leq c d_H(\Phi(0), \Phi) \gamma^i$$

in which

$$\begin{aligned} \Phi(i) &= \tilde{A} \Phi(i-1) + \tilde{B} \Psi \\ \Phi &= \tilde{A} \Phi + \tilde{B} \Psi \end{aligned}$$

and $\tilde{A}$ is a stable matrix ($\rho(\tilde{A}) < 1$).

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